



## **ONE SEMESTER PROJECT**

### **PROJECT PROPOSAL**

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**BDS 3106**

**BACHELOR OF SCIENCE IN DATA SCIENCE**

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**PROJECT TITLE : SMART BANKLYTICS**

An Intelligent Data – Driven Bank Management and Customer Analytics System

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## 1. Introduction

The rapid growth of digital banking has led to an increase in financial transactions, generating large volumes of data daily. Managing, analyzing, and securing this data has become a critical challenge for financial institutions. Traditional banking systems focus mainly on transaction processing, offering limited capabilities for data analysis and fraud detection.

This project aims to design and develop **Smart Banklytics**, an intelligent banking system integrated with data analytics and machine learning techniques. The system will provide standard banking functionalities while leveraging data science to generate insights, detect fraudulent transactions, and support decision-making. The project seeks to bridge the gap between traditional banking systems and modern data-driven financial platforms.

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## 2. Background Study

The rapid advancement of digital banking and financial technology has significantly transformed how financial transactions are conducted. Modern banking systems now process large volumes of electronic transactions daily, generating extensive datasets that can provide valuable insights into customer behavior, transaction trends, and overall financial performance. This shift has increased the demand for intelligent systems capable of managing, analyzing, and securing financial data efficiently.

Despite these advancements, many small-scale banking systems and educational banking applications primarily focus on basic transaction processing such as deposits, withdrawals, and fund transfers, with limited emphasis on analytics and fraud detection. As a result, administrators often lack effective tools for monitoring financial activities, identifying suspicious transactions, and generating meaningful insights from transaction data.

With the growing importance of data science and machine learning, there is a need to develop intelligent banking systems that integrate transaction management with analytics and fraud detection. This project seeks to address these challenges by developing Smart Banklytics, a data-driven banking analytics and fraud detection system that enhances transparency, security, and decision-making.

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### **3. Problem Statement**

Many small and medium-scale banking systems lack integrated analytics and fraud detection mechanisms. As a result, financial institutions face challenges in monitoring customer behavior, detecting suspicious activities, and making informed financial decisions. The absence of real-time data analysis and predictive capabilities increases the risk of fraud, poor financial planning, and inefficient service delivery. This project addresses these challenges by developing an intelligent banking system that integrates transaction processing, analytical dashboards, and automated fraud detection models to enhance efficiency, security, and decision-making.

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### **4. Objectives**

#### **Main Objective**

To develop an intelligent banking system that integrates transaction management, data analytics, and fraud detection using data science techniques.

#### **Specific Objectives**

1. To review and analyze transaction data to generate insights and detect suspicious activities.
  2. To design and implement a secure banking system for user account management and financial transactions.
  3. To collect and store transaction data for analytical processing.
  4. To develop an interactive analytics dashboard for transaction trend visualization.
  5. To build a machine learning-based fraud detection model for identifying suspicious transactions.
  6. To generate automated transaction reports and downloadable bank statements.
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### **5. Literature Review**

Several studies have explored the development of digital and web-based banking systems that enable users to perform essential financial operations such as deposits, withdrawals, and fund transfers. These systems have improved accessibility and efficiency by allowing users to conduct banking activities remotely. However, most of these systems primarily emphasize transaction

processing and account management, offering limited capabilities for advanced data analytics and decision support.

Recent research highlights the importance of integrating data analytics and visualization tools within banking systems to support financial monitoring, customer behavior analysis, and strategic decision-making. Additionally, machine learning techniques such as anomaly detection and classification algorithms have been widely applied to detect fraudulent transactions. While these approaches have proven effective, their implementation is often absent in small-scale or educational banking platforms due to technical complexity and limited system design scope.

From the reviewed literature, it is evident that a significant gap exists between traditional transaction-based banking systems and intelligent data-driven platforms. This project aims to bridge this gap by integrating transaction management, analytics dashboards, and automated fraud detection mechanisms into a unified system, thereby enhancing operational efficiency, transparency, and security.

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## **6. Scope of the Project**

The project will focus on:

- User account registration and authentication.
- Deposit, transfer and withdrawal processing.
- Transaction history tracking.
- Data storage and management.
- Analytical dashboard visualization.
- Fraud detection using machine learning.
- PDF transaction statement generation.

The system will be implemented as a data-driven banking management and analytics platform with a web-based interface for interaction, visualization, and reporting.

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## **7. Methodology**

## System Development Approach

The project will follow an **Agile development methodology**, enabling incremental development, testing, and improvement of system modules.

## System Architecture Overview

- i. **Frontend Interface:** User interaction layer for banking operations.
  - ii. **Backend System:** Handles business logic and transaction processing.
  - iii. **Database:** Stores user details and transaction records.
  - iv. **Analytics Engine:** Processes transaction data for insights.
  - v. **Machine Learning Module:** Detects fraudulent transactions.
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## Implementation Steps

- a) **Requirement Analysis**
  - a. Identify system features and functional requirements
- b) **System Design**
  - a. Database schema design
  - b. User interface design
  - c. System workflow planning
- c) **System Development**
  - a. Backend implementation using Flask
  - b. Database integration
  - c. Transaction management logic
- d) **Data Analytics Development**
  - a. Data cleaning and preprocessing
  - b. Trend analysis and visualization
  - c. Dashboard implementation

e) **Machine Learning Model Development**

- a. Data preparation
- b. Model training using Scikit-learn
- c. Fraud prediction integration

f) **Testing and Evaluation**

- a. Functional testing
- b. Model performance evaluation

g) **Deployment and Documentation**

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**Tools and Technologies**

- **Programming Language:** Python
- **Backend Framework:** Flask
- **Database:** SQLite / MySQL
- **Data Analysis:** Pandas, NumPy
- **Visualization:** Matplotlib, Plotly, Streamlit
- **Machine Learning:** Scikit-learn
- **Report Generation:** ReportLab / WeasyPrint (PDF generation)

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**7. Expected Outcomes**

- A fully functional intelligent banking system
- Secure transaction processing environment
- Downloadable customer bank statements
- Interactive transaction analytics dashboard
- Automated fraud detection mechanism
- Improved financial insights and reporting

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## **8. Project Significance**

The project will demonstrate the practical application of data science concepts in financial systems. It will enhance understanding of transaction analytics, fraud detection, and data-driven decision-making. The system can be adapted for educational use, financial research, and prototype banking solutions.

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## **9. Conclusion**

Smart Banklytics aims to combine banking operations with intelligent data analysis and fraud detection to create a modern, efficient, and secure financial system. The project will showcase the integration of software engineering and data science principles, providing a strong foundation for real-world financial system development.